1 Absolute value on an ordered field

Definition 1.1. Let F be an ordered field (not necessarily complete). The **absolute value** of $x \in F$ is

$$|x| := \begin{cases} x, & \text{if } x \geq 0 \\ -x, & \text{if } x < 0. \end{cases}$$

Remark 1.1. In the above definition we write 0 for $\mathbf{0}$ (the additive identity of the field F), and $-x$ stands for the additive inverse of x.

Remark 1.2. $|x| \in F \; \forall x \in F$.

Properties

Let F be an ordered field. Then $\forall x, y, a \in F$:

(i) $|x| \geq 0$; and $|x| = 0$ if and only if $x = 0$.

(ii) $|x| = |-x|$.

(iii) $-|x| \leq x \leq |x|$.

(iv) $|x| \leq a$ if and only if $-a \leq x \leq a$. Similarly, $|x| < a$ if and only if $-a < x < a$.

(v) $|x \otimes y| = |x| \otimes |y|$.

Proof. We prove (iii) and (iv); the rest is an **exercise**.
(iii) If $x \geq 0$, then $|x| = x$ and $-|x| = -x \leq 0 \leq x$. If $x < 0$, then $|x| = -x > 0 > x$ and $-|x| = x$. Either way, $-|x| \leq x \leq |x|$.
(iv) ($\Rightarrow$) If $|x| \leq a$, then by (iii), $x \leq |x| \leq a$ and $-a \leq -|x| \leq x$. So we get $-a \leq x \leq a$.
($\Leftarrow$) Suppose $-a \leq x \leq a$. If $x \geq 0$, then $|x| = x \leq a$; if $x < 0$, then $|x| = -x \leq a$ since $-a \leq x$. In either case, $|x| \leq a$.

□

Triangle inequality

Theorem 1.1 (Triangle inequality). Let F be an ordered field, and $x, y \in F$. Then

$$|x \oplus y| \leq |x| \oplus |y|. \tag{*}$$

Proof of (*). Adding the inequalities $-|x| \leq x \leq |x|$ and $-|y| \leq y \leq |y|$ from (iii), and using $(-|x|) \oplus (-|y|) = -(|x| \oplus |y|)$ (distributivity with $-x = (-1) \otimes x$), we get

$$-(|x| \oplus |y|) \leq x \oplus y \leq |x| \oplus |y|.$$

Now apply (iv) with $a = |x| \oplus |y|$. □

2 Metrics and topology

Definition 2.1. Let S be any set. A function $d : S \times S \to \mathbb{R}$ is called a **metric** or **distance** on S if $\forall x, y, z \in S$ the following three properties hold:

(i) $d(x, y) \geq 0$, and $d(x, y) = 0$ if and only if $x = y$ — *positivity.*

(ii) $d(x, y) = d(y, x)$ — *symmetry.*

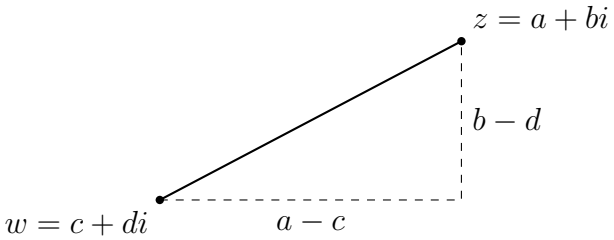
(iii) $d(x, z) \leq d(x, y) + d(y, z)$ — *triangle inequality.*

Example 2.1. In $\mathbb{R}$, $d(x, y) = |x - y|$ is a metric, called the *usual metric* on $\mathbb{R}$. Positivity follows from property (i) of the absolute value, symmetry from property (ii) (since $y - x = -(x - y)$), and the triangle inequality from (*), since:

$$d(x, z) = |(x - y) + (y - z)| \leq |x - y| + |y - z| = d(x, y) + d(y, z).$$

Example 2.2. In $\mathbb{C}$, $d(z, w) = |z - w|$ is a metric (a distance) on $\mathbb{C}$; we will verify this in Section 3 (see Proposition 3.1). Writing $z = a + bi$ and $w = c + di$,

$$d(z, w) = \sqrt{(a - c)^2 + (b - d)^2}.$$



Definition 2.2. Let S be a set, and d a metric (a distance) on it. The **open ball** centered at $z \in S$ with radius $r \in \mathbb{R}$, $r > 0$, is the following subset of S:

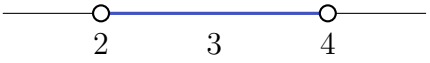
$$B(z, r) := \{w \in S : d(w, z) < r\}.$$

Remark 2.1 (Notation). The open ball depends on the metric, not just on the set S: the same set can carry several different metrics, and each gives different balls (see the end of this section). When we need to make the metric clear, we write it as a subscript, as in $B_d(z, r)$, and similarly $D_d(z, r)$ for closed balls. When only one metric is in use, such as the usual metric on $\mathbb{R}$ or $\mathbb{C}$, we simply write $B(z, r)$.

Example 2.3.

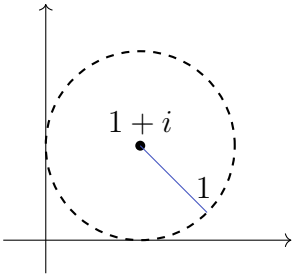
(i) If $S = \mathbb{R}$ with $d(x, y) = |x - y|$, then what we have just called an open ball is an open interval. For example, in $\mathbb{R}$ we have $B(3, 1) = (2, 4)$:

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(ii) Here is a picture of the open ball of radius 1 in $\mathbb{C}$ centered at $1 + i$:

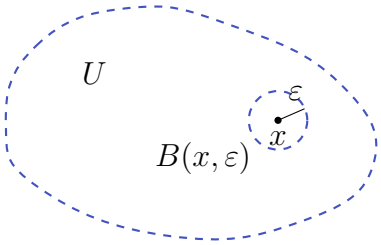
$B(1+i,1)$:



Definition 2.3. A subset $U \subseteq S$ is *open* if it contains an open ball around each of its points. This means that for all $x \in U$, there exists $\varepsilon > 0$ such that

$$B(x, \varepsilon) \subseteq U.$$

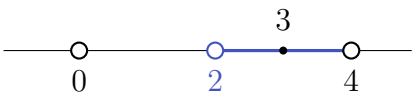
A key point here is that the open ball must be *entirely contained* inside U:



In the above picture, $U \subset \mathbb{C}$ is the set of all points inside the outer dashed line. Inside U, we show a typical point x and an example of an open ball around x completely contained inside U. As we test points nearer the boundary of U, the choice for ε will need to become smaller.

Example 2.4.

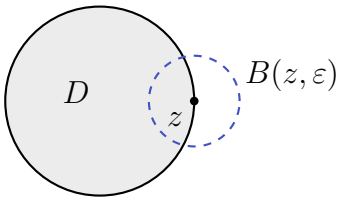
(i) An open interval $(a, b) \subseteq \mathbb{R}$ is open. Given $c \in (a, b)$, let ε be either $c - a$ or $b - c$, whichever is smaller. Then the open interval centered at c and with radius ε is contained in (a, b). For instance, take the interval $(0, 4)$ and let $c = 3$. Then $(0, 4)$ contains the open ball of radius 1 around c, i.e., it contains the interval $(2, 4)$.



(ii) A union of open intervals on $\mathbb{R}$ is an open set. In fact, every *open* subset of $\mathbb{R}$ is a union of open intervals. (See Proposition 2.1).

(iii) The half-open interval $[1, 2)$ is not open. The problem is the endpoint 1. We have $1 \in [1, 2)$, but there is no open ball (interval) centered at 1 and contained in $[1, 2)$. Any open interval around 1 will contain points less than 1 and hence not in the set $[1, 2)$.

(iv) The closed disc $D := \{z \in \mathbb{C} : |z| \leq 1\}$ is not open. The problem is each of the points with modulus 1 sitting on the boundary of D. Any open ball around one of these points will contain points that are not in D, i.e., it will not be completely contained in D:



Proposition 2.1. Every open set is a union of open balls.

Proof. Let $U \subseteq S$ be open. Since U is open, for each $u \in U$, there exists $\varepsilon_u > 0$ such that

$$B(u, \varepsilon_u) \subseteq U.$$

We claim that

$$\bigcup_{u \in U} B(u, \varepsilon_u) = U.$$

To see this, first let $v \in \bigcup_{u \in U} B(u, \varepsilon_u)$. Then $v \in B(u, \varepsilon_u)$ for some $u \in U$, and since $B(u, \varepsilon_u) \subseteq U$, we have $v \in U$. To see the reverse inclusion, let $v \in U$. Then $v \in B(v, \varepsilon_v)$ and hence v is in the union $\bigcup_{u \in U} B(u, \varepsilon_u)$. □

Proposition 2.2. Let d be a metric on S.

(i) A union of open sets (even infinitely many) is open.

(ii) A finite intersection of open sets is open.

Proof. **Exercise.** (For (ii), given $x \in U_1 \cap \cdots \cap U_n$, take ε to be the smallest of the radii that work for each U_i.) □

Remark 2.2. The sets $\varnothing$ and S are always open, and together with Proposition 2.2 this says that the open sets of a metric space form what is called a *topology* on S. You will learn more about topology in later courses.

Regarding part (ii), note that an infinite intersection of open sets might not be open. For example,

$$\bigcap_{n=1}^{\infty} \left(-\frac{1}{n}, \frac{1}{n}\right) = \{0\}.$$

Why is $\{0\}$ not open with respect to the usual metric on $\mathbb{R}$?

Definition 2.4. A subset $C \subseteq S$ is *closed* if its complement $S \setminus C$ is open.

Example 2.5.

(i) A closed interval $[a, b]$ in $\mathbb{R}$ is closed with respect to the usual metric. That's because its complement

$$\mathbb{R} \setminus [a, b] = (-\infty, a) \cup (b, \infty)$$

is open.

(ii) If $z \in S$, then the set $\{z\}$ is closed. Why?

Definition 2.5. The *closed ball (disc)* of radius $r > 0$ centered at $z \in S$ is

$$D(z, r) := \{w \in S : d(w, z) \leq r\}.$$

Example 2.6. In $\mathbb{R}$, closed discs are the same as closed intervals $[a, b]$ with $a < b$. What are the closed discs in $\mathbb{C}$?

Proposition 2.3. An open ball is open.

Proof. Let $B(c, r)$ be an open ball in S, and let $z \in B(c, r)$. We must show there is an open ball around z that is completely contained in $B(c, r)$. Let $\varepsilon := r - d(z, c)$, and note that $\varepsilon > 0$ since $d(z, c) < r$.
We claim that $B(z, \varepsilon) \subseteq B(c, r)$. To see this, let $w \in B(z, \varepsilon)$. Then by symmetry and the triangle inequality,

$$\begin{aligned} d(w, c) &\leq d(w, z) + d(z, c) \\ &< \varepsilon + d(z, c) \\ &= r. \end{aligned}$$

Thus $w \in B(c, r)$. We can conclude that $B(z, \varepsilon) \subseteq B(c, r)$. □

Here is a picture motivating the above proof:

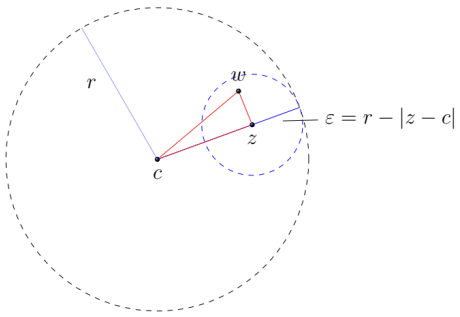
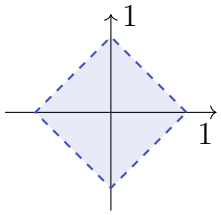


Figure 1: Picture motivating the above proof: an open ball is an open set.

Same points, different distances

On $\mathbb{R}^2$, the **Manhattan distance** $d_1\big((x_1,x_2),(y_1,y_2)\big) = |x_1-y_1|+|x_2-y_2|$ measures distance along horizontal and vertical streets. Its unit ball $B_{d_1}(0,1)$ is a diamond, unlike the round unit disc for the usual metric (compare Remark 2.1):



$$B_{d_1}(0,1) = \{(x,y) : |x|+|y|<1\}$$

Exercise 2.1 (The discrete metric). On any set S, the **discrete metric** is $d_{\text{disc}}(x,y) = 0$ if $x = y$ and $d_{\text{disc}}(x,y) = 1$ if $x \neq y$.

(i) Show that d_{disc} is a metric.

(ii) Describe $B_{d_{\text{disc}}}(x,r)$ for $0 < r \leq 1$ and for $r > 1$.

(iii) Show that every subset of S is both open and closed with respect to d_{disc}.

(iv) Compare $B_{d_{\text{disc}}}(x,1)$ with $D_{d_{\text{disc}}}(x,1)$.

Remark 2.3. In $\mathbb{R}$ with the usual metric, only $\varnothing$ and $\mathbb{R}$ are both open and closed (we'll see why later). So openness depends on the metric, not just the set.

3 Norms

Definition 3.1. Let F be any field. A function $\|\cdot\| : F \to \mathbb{R}$ is called a **norm** on F if $\forall x, y \in F$ the following three properties hold:

(i) $\|x\| \geq 0$, and $\|x\| = 0$ if and only if $x = \mathbf{0}$ — *positivity.*

(ii) $\|x \otimes y\| = \|x\| \cdot \|y\|$ — *multiplicativity.*

(iii) $\|x \oplus y\| \leq \|x\| + \|y\|$ — *triangle inequality.*

Example 3.1. Let $F = \mathbb{C}$. Then the modulus on $\mathbb{C}$ is a norm on $\mathbb{C}$.

Example 3.2. Let $F = \mathbb{R}$. Then the absolute value on $\mathbb{R}$ is a norm on $\mathbb{R}$. This follows immediately from properties (i) and (v) and the triangle inequality (*) in Section 1.

Proposition 3.1. Let F be a field, and $\|\cdot\|$ a norm on it. Then the function $d : F \times F \to \mathbb{R}$ defined by $d(x, y) = \|x \ominus y\|$ is a metric on F.

Proof. We need to show that $d(x, y) = \|x \ominus y\|$ satisfies the three properties from Definition 2.1.

(i) First note that $d(x, y) \geq 0$ follows immediately from the positivity of the norm. We also have that $d(x, y) = 0$ if and only if $\|x \ominus y\| = 0$ if and only if $x \ominus y = \mathbf{0}$, which holds if and only if $x = y$.

(ii) For symmetry, we first show $\|\mathbf{1}\| = 1$. Taking $x = y = \mathbf{1}$ in multiplicativity gives $\|\mathbf{1}\| = \|\mathbf{1} \otimes \mathbf{1}\| = \|\mathbf{1}\|^2$, and $\|\mathbf{1}\| \neq 0$ since $\mathbf{1} \neq \mathbf{0}$, so $\|\mathbf{1}\| = 1$. Moreover, $(-\mathbf{1}) \otimes (-\mathbf{1}) = \mathbf{1}$, and thus $\| - \mathbf{1}\|^2 = 1$, which implies by nonnegativity that $\| - \mathbf{1}\| = 1$. Hence

$$\|x \ominus y\| = \|(-\mathbf{1}) \otimes (y \ominus x)\| = \| - \mathbf{1}\| \cdot \|y \ominus x\| = \|y \ominus x\|,$$

and symmetry follows.

(iii) Finally we show the triangle inequality. For $x, y, z \in F$,

$$d(x, y) = \|x \ominus y\| = \|(x \ominus z) \oplus (z \ominus y)\| \leq \|x \ominus z\| + \|z \ominus y\| = d(x, z) + d(z, y). \quad \square$$

In particular, combining Proposition 3.1 with Example 3.1 shows that $d(z, w) = |z - w|$ is a metric on $\mathbb{C}$, as claimed in Example 2.2.